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Using High-Resolution Climate Models to Identify Climate Change Hotspots in the Middle East: A Case Study of Iran

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Abstract: The adverse effects of climate change will impact all regions around the world, especially Middle Eastern countries, which have prioritized economic growth over environmental protection. However, these impacts are not evenly distributed spatially, and some locations, namely climate change hotspots, will suffer more from climate change consequences. In this study, we identified climate change hotspots over Iran—a developing country in the Middle East that is facing dire economic situations—in order to suggest pragmatic solutions for vulnerable regions. We used a statistical index as a representative of the differences in climatic parameters for the RCP8.5 and RCP4.5 forcing pathways between historical data (1975–2005), near-future data (2030–2060) and far-future data (2070–2100). More specifically, we used downscaled high-resolution (0.25°) meteorological data from five GCMs of the CMIP5 database to calculate the statistical metric. Results indicate that for the far-future period and RCP4.5, regions stretching from the northwest to southeast of Iran, namely the Hotspot Belt, are the most vulnerable areas, while, for RCP8.5, almost the whole country is vulnerable to climate change. The highest and lowest differences in temperature for RCP8.5 in 2070–2100 are observed during summer in the northwestern and central parts and during winter in the northern and northeastern parts. Moreover, the maximum increase and decrease in precipitation are identified over the western parts of Iran during fall and winter, respectively. Overall, western provinces (e.g., Lorestan and Kermanshah), which are mostly reliant on rainfed agriculture and other climate-dependent sectors, will face the highest change in climate in the future. As these regions have less adaptive capacity, they should be prioritized through upstream policy change and special budget allocation from the government to increase their resiliency against climate change.

Keywords: climate change hotspots; climate models; RCPs; Middle East; Iran

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1. Introduction

By the end of this century, people in both poor and wealthy nations will be exposed to the relative challenges of climate change [1–4]. The increase in the annual mean Earth temperature and in the frequency and magnitude of extreme events such as floods, droughts, dust storms, sea level rise and arctic sea ice decline are all caused or intensified by climate change [5,6]. Moreover, climate change can threaten water and food security to the extent that, by 2050, 70% of cropland areas will need more water to sustain the same level of food production [7]. Although it is firmly established that climate change is caused by human activities [2,8], it is still a controversial issue in international politics, especially in Middle Eastern countries, where economic growth has higher priority over emission reduction and climate change adoption policies [9]. Based on a NASA report, the average temperature of the Earth has risen by nearly 0.8 °C during the last century [10]. Moreover, due to the reduction in glacier volume in polar regions by around 13.1% each decade, the global mean

sea level has risen by approximately 3.3 mm annually [11]. An Intergovernmental Panel on Climate Change (IPCC) report shows that by continuing under the current condition and considering different emission scenarios and simulation models, the Earth's temperature may increase by between 1.8 and 3.7 °C by the end of the 21st century [12]. Similarly, historical data and numerous developed models show the adverse impacts of climate change on human health [13], animals [14] and natural sources [10]. Therefore, investigations on climate change impacts on a regional scale have become a focus of research in recent years, and the implementation of mitigation and adoption policies to achieve climate resilience has received more support.

Middle Eastern countries represent around a 7% contribution to GHG emissions on the global scale. More specifically, 60% of the total GHG emissions come from four countries, namely Iran, Kuwait, Saudi Arabia and the UAE [15]. The IPCC reports that the average temperature rise in the Middle East could be up to 2 and 4 °C in the next 20 years and by the end of the 21st century, respectively [1], which will be more than the global average temperature rise (1.8–3.7 °C). Iran is the second-largest nation in the Middle East, which, in terms of CO_2 cumulative production, occupies the first rank in this region and seventh in the world, with nearly 170 million tons of carbon [16]. UNFCCC estimates that Iran may experience an average increase in temperature of around 2.6 °C (more than the Middle East's average temperature rise) and a 35% reduction in precipitation by 2030 [17]. Moreover, in recent years, there has been an increase in the frequency and magnitude of natural disasters in Iran [18]. The series of devastating flash floods in March 2019 and July 2022 that inundated multiple cities in Iran is an example of these extreme events [19,20]. In addition, Iran's groundwater resources have been significantly depleted in the past few decades. A recent study by Ashraf et al. [21] shows that almost all the basins in Iran have been depleted at a rate between 20% and 2600% in the course of 14 years (2002–2015). Another study based on satellite gravimetry observations (GRACE mission) showed that Iran lost 211 ± 34 km^3 of its total water storage between 2003 and 2019, which is twice as much as Iran's annual water consumption, mainly due to extensive and unsustainable human water withdrawal, which is coupled with the climate-driven prolonged drought starting from the early twenty-first century [22]. Moreover, in some major water bodies, such as Urmia Lake and the Shadegan international wetlands, water levels have significantly declined, mainly due to climatic change and unsustainable water management [23,24]. On the other hand, major local and regional dust sources, as a component of human activities and climate change effects, have impacted different parts of Iran, with more frequent dust storms across various episodes each year. In this regard, many populated cities have experienced a high PM concentration in recent years [6,23,25,26].

According to previous studies, without any prevention, mitigation and adaptation policies, Iran will be vulnerable to climate change consequences that will severely harm residents, infrastructure and environmental resources [18]. Although climate change is a global and regional phenomenon [1,9,12,18,27], people in developing countries are disproportionately vulnerable to its effects. Therefore, it is necessary to recognize the response to global warming at a local sub-country scale, where the impact is more strongly felt [28], using accurate high-resolution climatic data. While general adoption policies are in place, they should be implemented at a local scale, where the adoptive capacity is usually lower, and regulations must be modified accordingly. In this regard, the first step is to detect the area that has a stronger response to climate change. In this study, we call these regions climate change hotspots (CCH). A CCH is defined by aggregating the distance between multiple climate variables in the past and future (the formulation will be discussed further in Section 2.3). The aggregated new metric could identify the potential areas that will face the greatest change in the climate. Detecting hotspots is more crucial for developing countries such as Iran, with a low to average GDP per capita, aging infrastructure and old industries with high emissions of GHGs. In fact, the effectiveness of climate change adaptation strategies depends largely on the identification of potential hotspots [3].

The literature on identifying climate change hotspots is abundant and is still growing. During the last decade, many studies have proposed different analytical methods to identify climate hotspots in the future [2,9,29–34]. For instance, Giorgi [35] defined a Regional Climate Change Index (RCCI) based on the change and interannual variability in the air temperature and precipitation data to identify the climate change hotspots globally. The author specified the Mediterranean, Northeastern European and Northern Hemisphere high-latitude areas and Central America as the major climate hotspots. In another major study, Williams et al. [29] suggested the Standardized Euclidean Distance (SED) to quantify differences between past and future conditions and detect regions that are projected to experience a high magnitude of local climate change. They used multiple models under the A2 and B1 emission scenarios from the Fourth Assessment Report (AR4) of the IPCC to identify areas that will face the most severe consequences of climate change. The probability of new climate formation and the disappearance of the existing climate was also examined at the end of the 21st century. In some other studies [2,3,9,36], the statistical representative metric was developed and different climate indices were used to increase the accuracy of hotspot identification. Moreover, climatic data and general circulation models (GCMs) from new versions of the CMIP protocol and updated emission scenarios—according to AR5 of the IPCC—have been applied to identify hotspots more accurately [36,37].

Over the past few decades, some investigations have been performed over Iran using GCMs with a focus on water resource variation due to climate change and the frequency and intensity of extreme events, such as droughts, heat waves, precipitation and flood frequency, in the future [18,38–40]. However, to date, no research seems to have studied climate hotspot identification on a fine spatial scale (i.e., Iran). In this regard, we used five GCMs to compare meteorological parameters between the past and two timespans in the future and calculate the SED criterion to identify climate hotspots by considering two emission scenarios (i.e., RCP4.5 and RCP8.5), according to the Fifth Assessment Report (AR5) of the IPCC, with a focus on Iran. Most similar previous studies [3,9,29,36,37] were performed to identify future hotspots using statistical metrics on a global scale, with a coarse spatial resolution ($\geq 1^{\circ}$), which may not be suitable for regional and local scales. Therefore, in this study, we used high-resolution climatic data to calculate different statistical metrics.

Overall, the aim of this study is to aggregate the differences between temperature and precipitation as a unique representative metric to identify potential climate change hotspots. Meteorological data from five GCMs from the CMIP5 database with a fine spatial resolution are used to calculate the SED score. In Section 2, the study area is introduced and the input climatic data, the GCMs and the SED formulation are explained. The identified hotspots over Iran and the variations in five climatic indicators under the forcing pathways in the future are presented in Section 3. Finally, in Section 4, the obtained results are discussed and compared with similar previous studies in detail.

2. Method

2.1. Study Area

Iran, with an approximately 1650,000 km^2 area, is the second-largest nation in the Middle East, which is extended from the Caspian Sea to the Persian Gulf (Figure 1). Iran's altitude varies between −28 m on the southern coast of the Caspian Sea and 5610 in Mount Damavand (which are not far away from each other). Almost half of Iran belongs to the highlands, and the rest is equally divided between deserts and arable land [41]. According to the World Bank, in 2017, the average precipitation over Iran was 228 mm which is one fifth of the average annual precipitation over the globe. Moreover, because of the Alborz Mountains in the north and the Zagros Mountains in the west and southwest, Iran has a diverse climate. These various climate types include (1) humid–mild in the north, (2) a humid–cold climate in regions near the Alborz and Zagros mountains, (3) dry–cold climates in some central and northeastern parts, (4) humid–hot in the south and (5) a dry–hot climate for other parts [18]. Due to this large climatic variability, the temperature can

vary between −20 and +80.83 (the hottest temperature recorded on Earth was in the Lut desert) [42].

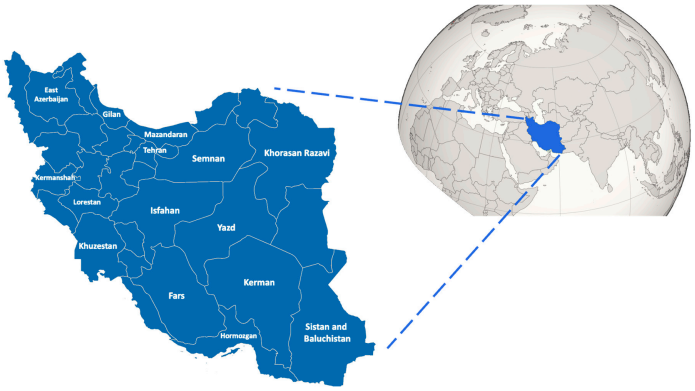


Figure 1. The location of the study area—Iran and its provinces.

2.2. Climatic Data and Models

As some previous studies have shown, changes in temperature and precipitation and an increase in the frequency and magnitude of extreme events (e.g., floods and droughts) could describe the probable change in climate in the future [9,18,29,37]. With the advances in climate modeling, many studies have used the capabilities of these models to identify climate-change-vulnerable regions [43,44]. Here, meteorological gridded data simulated by five GCMs, namely GFDL-ESM2M, CCSM4, IPSL-CM5A-LR, MIROC5 and NorESM1-M, from the CMIP5 archive, were used. All simulated datasets include two future timespans (2030–2060 and 2070–2100) and a baseline timespan (1975–2005), which were studied for two general Representative Concentration Pathways (RCPs)—RCP4.5 and RCP8.5—to identify climate-change-vulnerable regions in the future. Moreover, it should be noted that the bias in raw data has been removed.

In contrast to previous studies, which used coarse-resolution (e.g., >1 degree) climatic data, in this study, we used a new global high-resolution (0.25 degree) dataset from the NASA Earth Exchange Global Daily Downscaled Projections (NASA NEX-GDDP https://portal.nccs.nasa.gov/ (accessed on 31 October 2020) (the data used were available until 2021)). This dataset is produced by running multiple GCMs derived by two greenhouse gas emission scenarios, RCP4.5 and RCP8.5, that are conducted under the Coupled Model Intercomparison Project Phase 5 (CMIP5). Outputs of these models (air temperature and precipitation) have been downscaled and bias-corrected using synoptic observations and satellite data. The NEX-GDDP collection includes 21 climate models and has 56 years of historical data (1950–2005) along with climate projections for two RCPs for the 2006–2100 timespan.

2.3. Hotspot Formulation

The approach that was proposed by Diffenbaugh et al. [37] has been used to formulate climate change indices and identify major CCHs in Iran. In this method, the Standard Euclidean Distance (SED) is defined to compare the seasonal change between climate variables in a baseline and future periods. Based on the SED formula (Equation (1)), the different types of variabilities in mean values and the frequency of extreme climatic conditions for climate parameters (i.e., temperature and precipitation) were calculated between the present and future in each land grid cell.

$$SED_{total} = \sqrt{\sum_{i}^{N_\vartheta} \sum_{j}^{4} \left(\frac{|\Delta_\vartheta|}{max\ |\Delta_\vartheta|_{ij}} \right)^2} \quad (1)$$

In Equation (1), $|\Delta_\vartheta|$ is the absolute value of change in climate parameter ϑ in each grid cell between the baseline and future periods. Moreover, $max\ |\Delta_\vartheta|_{ij}$ is the maximum

absolute value of change in climate parameter ϑ for all land grid cells ij in the study area for the worst-case climate scenario (i.e., RCP8.5). This element (i.e., $max\ |\Delta_\vartheta|_{ij}$) was used as a denominator in the SED formula to normalize SED values in all cases to be able to compare the results during different periods. Moreover, the SED formula considers an equal coefficient for all climatic indices and was calculated for each GCM. In the next step, calculated SED scores were aggregated for all models to report a unique value for each grid cell. Overall, higher SED scores show more significant climate change impacts in the study region.

Using climate parameters (ϑ) including temperature and precipitation, five climate indicators were computed for each grid cell over the study area for each season (December–January–February (DJF), March–April–May (MAM), June–July–August (JJA) and September–October–November (SON)) between future (2030–2060 and 2070–2100) and baseline (1975–2005) timespans. These climate indicators are (1) the absolute change in maximum observed temperature $|\Delta T|$, (2) the percentage change in mean precipitation with respect to the mean over the baseline $\left|P_{frac}\right|$, (3) the frequency of seasons in which the maximum temperature was above the baseline maximum seasonal temperature f_{hot}, (4) the frequency of seasons in which the precipitation was below the baseline minimum seasonal precipitation f_{Dry} and (5) the frequency of seasons in which the precipitation was above the baseline maximum seasonal precipitation f_{wet}. Five climate indicators for four seasons led to the comparison of past and future climate conditions in 20 different dimensions. In this regard, 20 different values of Δ_ϑ were calculated for the baseline and future period for each cell and then aggregated to obtain the final value of SED for each cell. Additional details about hotspot identification steps are provided in Figure 2.

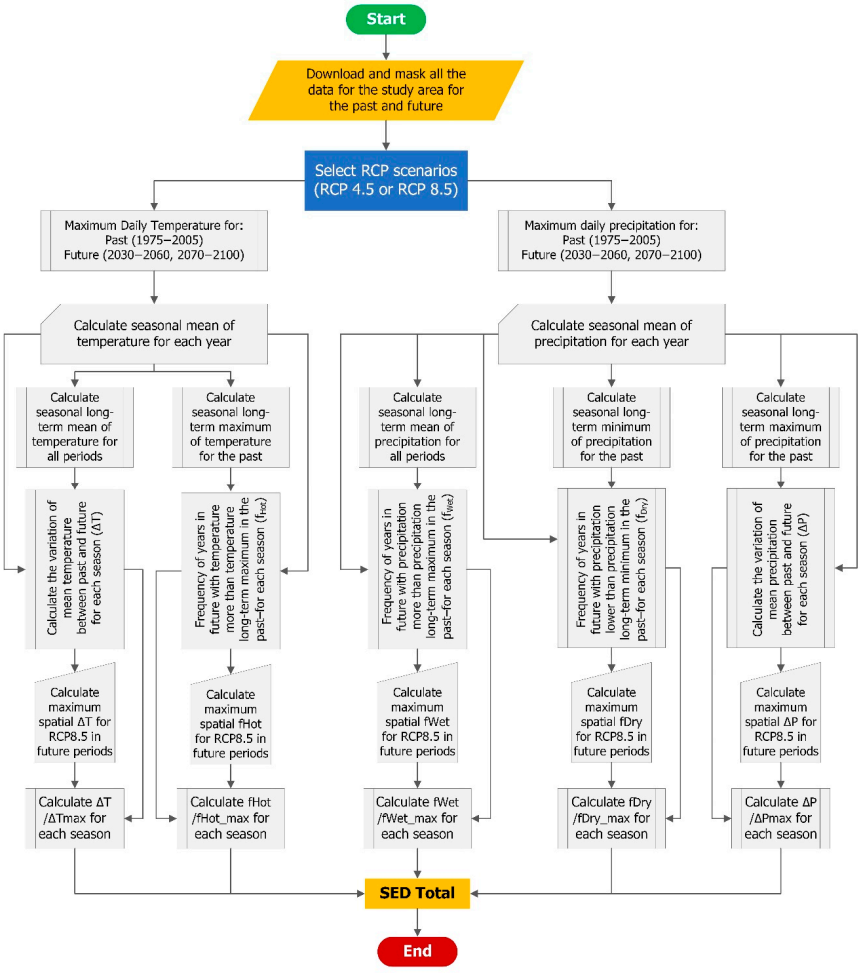


Figure 2. Hotspot identification flowchart.

3. Results

3.1. SED Variations

Figure 3 shows the climate change hotspot patterns obtained from the aggregation of five models for the two forcing pathways during two future timespans. SED scores for RCP4.5 in the period of 2030–2060 are indicated in Figure 3a. Clearly, climate change will impact the western and southern parts of Iran, with highly localized effects on Lorestan, Hormozgan and surrounding provinces. Climate change effects are less pronounced for the northern and eastern regions of Iran, with SED scores lower than 1.5. During the 2070–2100 period and in RCP4.5 (Figure 3b), all parts of Iran except a few regions in the north and northeast will be affected. The intensity of the climate change impact in this period (i.e., far future 2070–2100) is more than that in the near future (i.e., 2030–2060), with a greater spread over the entire study area. The region stretching from the southeast to northwest of Iran, which is named the *Hotspot Belt* henceforth, has the highest SED scores in RCP4.5, indicating that this is the most vulnerable region in Iran.

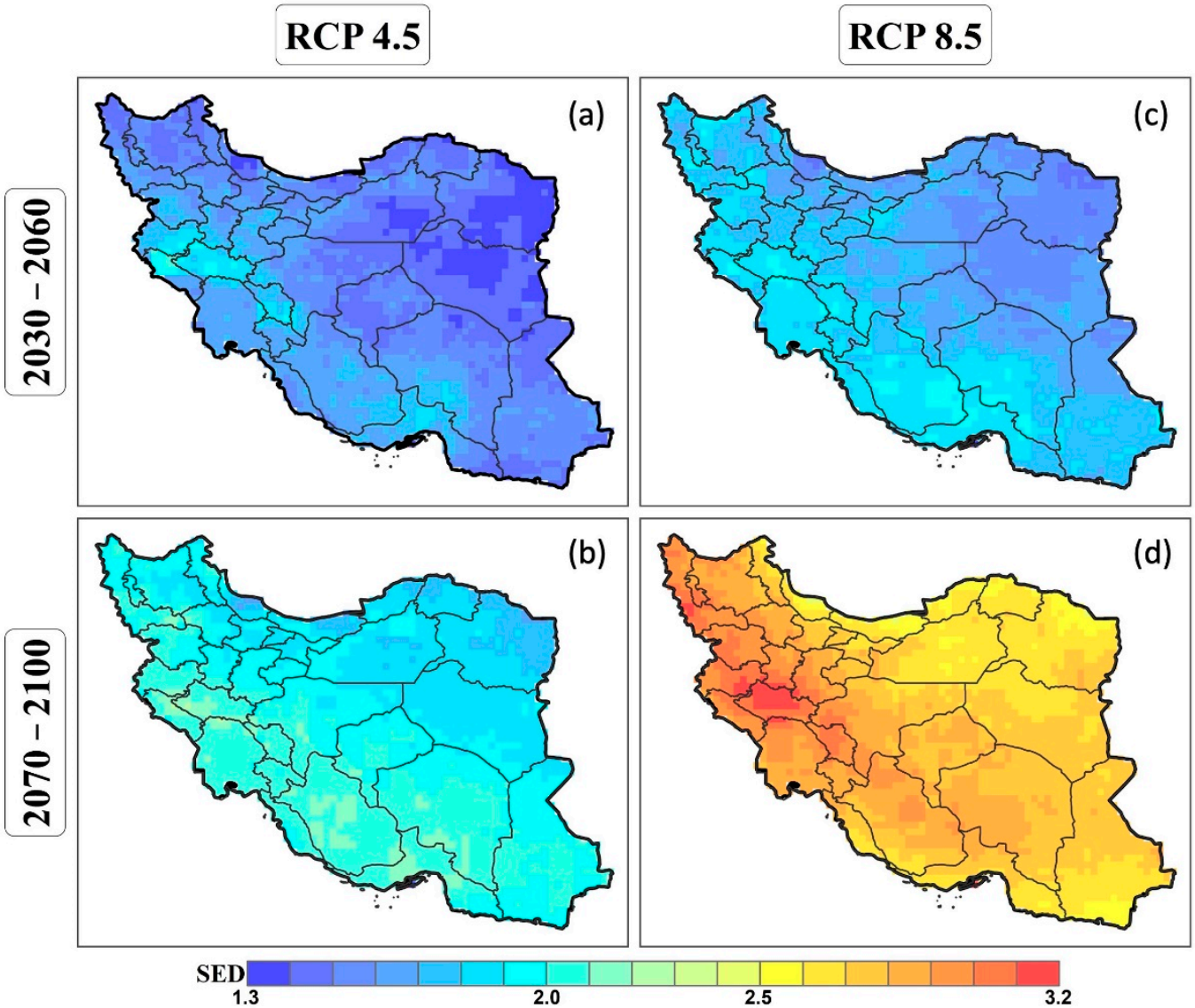


Figure 3. Hotspot pattern over Iran for the 2030–2060 and 2070–2100 periods under RCP4.5 (**a**,**b**) and RCP8.5 (**c**,**d**). Unitless.

The SED score map for RCP8.5 and the period 2030–2060 (Figure 3c) shows a similar pattern as that for RCP4.5 (same period), with a relatively intensified climate change

impact over the northwestern and western parts of Iran. In RCP8.5 and the far-future period (Figure 3d), the climate will change more strongly relative to the near-future period. Moreover, a generally similar pattern as the far future in RCP4.5 was repeated, with higher SED scores over the study area. All parts of the study area experience a significant level of climate change, especially over the *Hotspot Belt*, with a particular focus on the western parts of Iran (Lorestan province). In addition, the central parts of the study region show an intermediate intensity of climate change, while the minimum SED scores occur over the northern and northeastern areas during the 21st century. Overall, the peak level of climate change occurs in the late 21st century for RCP8.5, and the intensity of the observed pattern gradually decreases throughout the study area for the near-future period and RCP4.5.

Seasonal variations in SED for RCP8.5 during the two future periods are shown in Figure 4. During 2030–2060, summer and fall are the critical seasons in terms of climate hotspots, with the highest SED scores. In contrast, the lowest level of climate change may occur during winter. In the period of 2070–2100 as the worst-case scenario, SED scores peak in almost all seasons, especially during fall, with localized peaks over western and northwestern provinces. As observed previously in the yearly analysis, the *Hotspot Belt* pattern has been also repeated in the seasonal SED score maps.

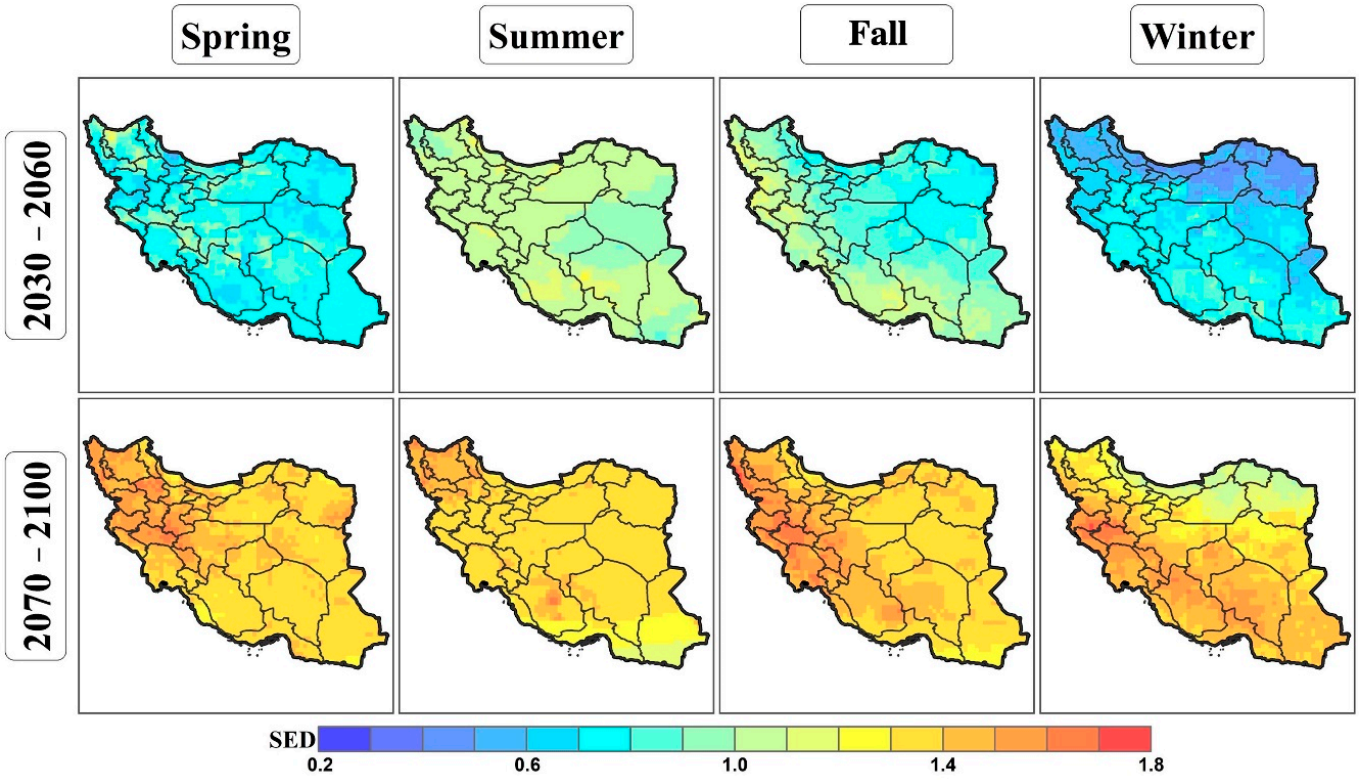


Figure 4. SED scores over the study area during the 2030–2060 and 2070–2100 periods for RCP8.5 during different seasons. Unitless.

3.2. Variations in Climate Indicators

The seasonal hotspot pattern for five climate change indicators related to the worst forcing pathway during 2070–2100 is represented in Figure 5. Generally, different parts of the *Hotspot Belt* have peak SED scores in most of the indicators. The seasonal temperature difference (ΔT) (Figure 5a) is projected to change significantly between 5.5 °C and 6.5 °C in most parts of Iran during summer (peak values in northwestern and central parts). In addition, the north and northeast of Iran experience the lowest variation in temperature during all seasons. The seasonal precipitation difference will strongly decrease during winter for the *Hotspot Belt*, especially in the western regions of Iran (i.e., $\Delta P = -60$ mm) (Figure 5b). Changes in precipitation are less pronounced in other seasons, except for the

slight increase during fall. Moreover, positive values of ΔP in the central and southern parts of Iran are observed in most seasons.

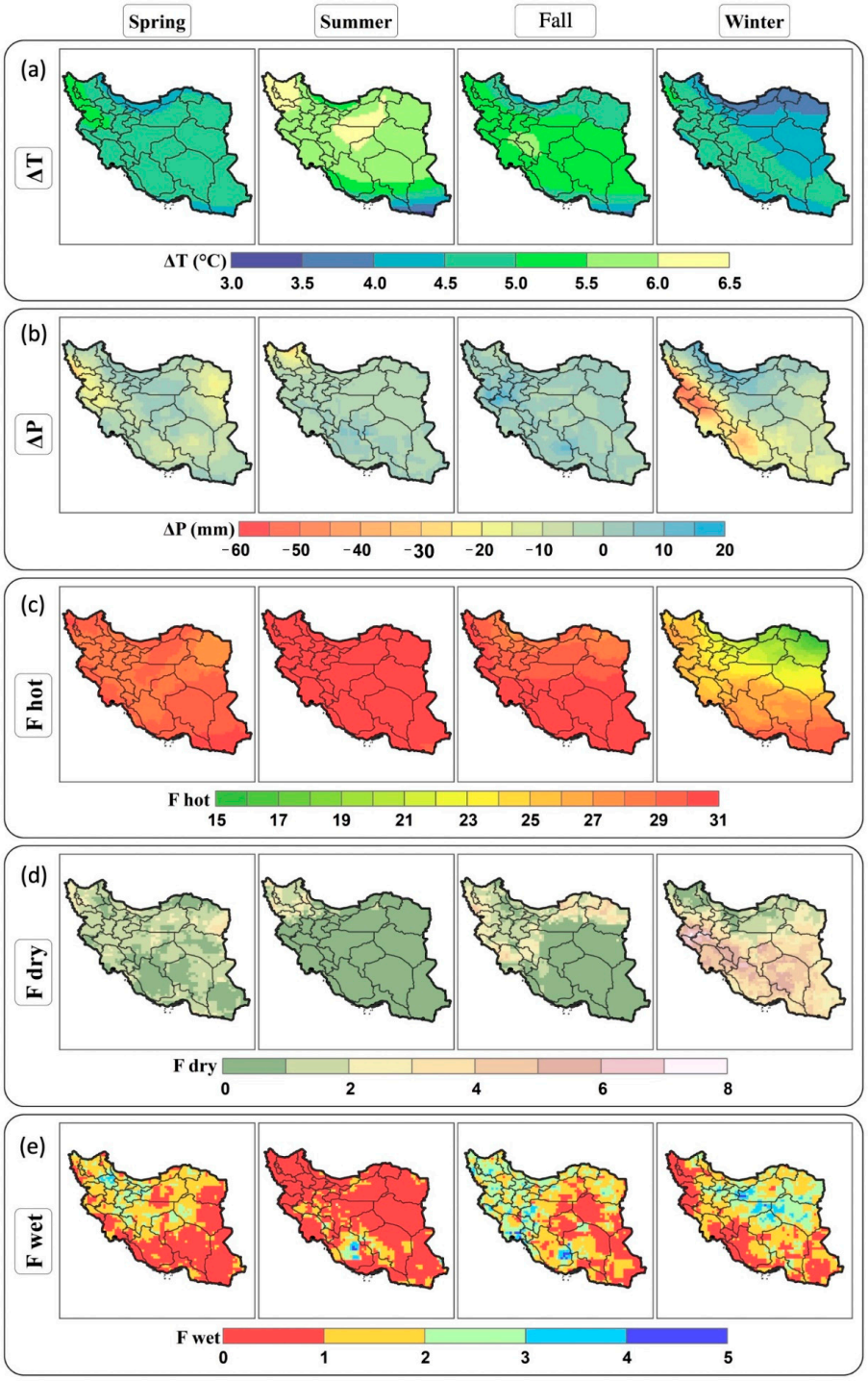


Figure 5. Variations in climate change indicators over Iran for RCP8.5 during 2070–2100 period (unitless). (**a**) Seasonal temperature difference (ΔT), (**b**) seasonal precipitation difference (ΔP), (**c**) seasonal temperature occurrences (F_{hot}), (**d**) frequency of extreme dry seasons (F_{Dry}) and (**e**) wet season frequency (F_{Wet}).

The frequency of extreme seasonal temperature occurrences (F_{hot}—Figure 5c) increases significantly during summer and winter in the range of 30–31 and 15–29 additional times, respectively, with the frequency in summer being nearly twice as high as in winter. All parts of Iran, especially the *Hotspot Belt* region, are expected to experience a high frequency

of extreme seasonal temperature occurrences during summer and winter. The frequency of extreme dry seasons (F_{Dry}—Figure 5d) has the maximum value of 8 during winter over the *Hotspot Belt*, but in the other seasons, it is homogenously low throughout the country. The extreme wet season frequency (F_{Wet}—Figure 5e) has the highest and lowest average over Iran during winter and summer, respectively. Wet seasons are predicted to occur mainly over the central part of Iran during winter and fall.

3.3. Climate Change over All Provinces of Iran

Due to the ratification of long-term upstream laws and regulations by the government in Iran, the identification of critical provinces and cities could help the government authorities to develop appropriate mitigation and adaptation strategies for future climate hotspots. This could be achieved by allocating a special annual budget for regions at risk to manage possible consequences of climate change. In this regard, average SED scores for each province were calculated for RCP4.5 and RCP8.5 during the 2030–2060 and 2070–2100 periods (Figure 6). For each timespan and RCP, the 80th percentile of SED scores was used to select critical provinces that may be impacted more than others due to climate change. In addition, the legend of Figure 6 is defined by the 80th percentile of SED scores for each of the four cases (i.e., 1.7, 1.8, 2 and 3). Therefore, the distance between the boundaries in the legend is not equal. According to Figure 6, for RCP4.5 and the near future, i.e., 2030–2060, Lorestan province, with an average SED score of 1.84, will be the most vulnerable province to climate change in the future. For RCP8.5 and the same period, eight more provinces will be impacted. For the far-future period, almost 40% and 100% of the provinces were identified as climate hotspots for RCP4.5 and RCP8.5, respectively. Lorestan and Kermanshah provinces in the western part of Iran are identified as the most vulnerable provinces to climate change.

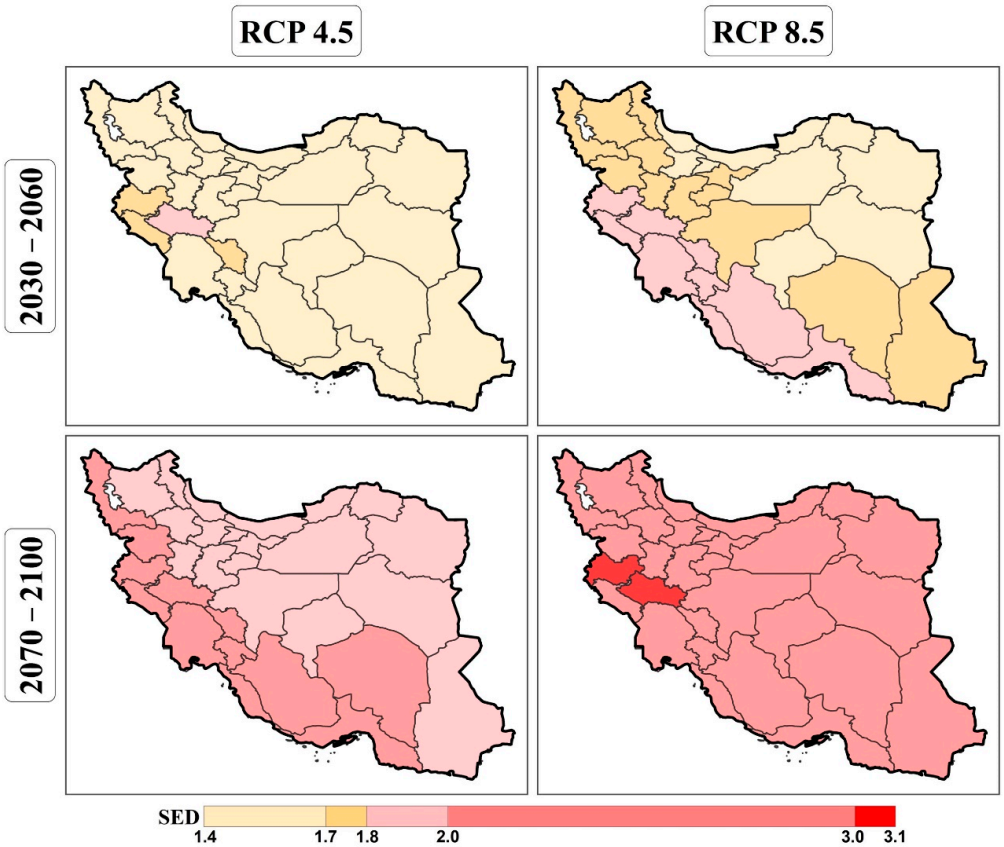


Figure 6. SED variations over all 31 provinces of Iran for RCP4.5 and RCP8.5 during 2030–2060 and 2070–2100 periods. Unitless.

4. Discussion and Conclusions

Due to the diversity of climatic regimes and the direct and indirect dependency of residents' livelihoods and family economy on climate in Iran, the identification of climatic hotspots in the future is crucial. According to these potentially vulnerable regions, adaptation and mitigation policies could be more effective and might help Iran's government to prepare for the consequences of climate change with a minimal level of damage. Therefore, in this study, we used representative statistical metrics along with high-spatial-resolution data to achieve our goals. Investigating climate change impacts for two emission scenarios in the near and far future indicates that, based on SED scores, the *Hotspot Belt* is the major climate change hotspot region over Iran. The comparison between hotspot regions identified in this study with other studies at a global scale [9,37] shows that our results are in agreement with previous studies. Their results indicate that the *Hotspot Belt* could be considered as the prominent hotspot over Iran, and then the vulnerable areas will gradually extend to other regions in the future.

In addition, the results related to seasonal temperature and precipitation differences (ΔT and ΔP) are consistent with similar previous studies that used the same indicators. In this regard, Diffenbaugh et al. indicated that, during summer, all parts of Iran would experience a high level of temperature; however, only the *Hotspot Belt* experiences the same situation during the winter. In addition, the climate change patterns during fall and spring are similar to those of summer and winter, respectively, which is in line with our results. For precipitation, their results agree with our study's results for summer and fall. During summer, the southern part would face positive rainfall, and during the fall, the western region of Iran will see an increase in precipitation. However, there are some differences between spring and winter [37]. As the results show, the northwestern parts, which have a humid–cold climate and high population density, will undergo a significant increase in temperature in the future that will affect and change the residents' lives and will be a driver of migration in these regions. Moreover, the southern part of Iran, on the coast of the Oman Sea, would experience the minimum ΔT in all seasons, which shows a stable climate condition in this region in the future. On the other hand, positive values of ΔP in the central and southern parts of Iran in most seasons could trigger flash floods, as has been observed more frequently in recent years [45], and impact the urban infrastructure drastically [5]. Moreover, the severe decrease in precipitation in the *Hotspot Belt*, especially the western parts, during winter could be a significant driver of drought issues in the future. To prevent the probable severe damage, governments and policymakers should establish some adaption policies and set up climate-resilient infrastructures in this region to deal with the climate change effects. The results obtained for F_{Hot} in the study by Diffenbaugh et al. [37] indicated that Iran would have similar patterns during all seasons except for winter, where this indicator shows a higher average frequency over the southern and western parts of Iran. Moreover, for F_{Dry}, the higher frequencies belong to winter over the *Hotspot Belt*, and other seasons have similar frequency patterns. Both frequency patterns for F_{Hot} and $F_{Dry,}$ in Diffenbaugh et al., are similar to our study. On the other hand, the comparison between our study and Diffenbaugh et al.'s result for F_{Wet} shows that, for all seasons except spring, the frequency patterns are similar. The main point is the prediction of areas with the highest frequency in the southwest of Iran, which is repeated in both studies [37].

In addition, based on a previous work [9], in tropical regions, temperature variation is much lower than at high latitudes; therefore, precipitation changes are the main driver of SED peak scores. In contrast, for the high-latitude area, temperature variations may be the prominent factor causing higher SED scores. Due to the type of climate condition of the study area, and the variation intervals of the five indicators (e.g., F_{Hot} (15–31), F_{Dry} (0–8) and F_{Wet} (0–5)), temperature is the main indicator that has significantly affected the SED scores. Although most of the results of our study are in agreement with previous studies, different sources of uncertainty could affect them. In this regard, the number and type of GCMs used in the ensemble, the model resolution, the selected meteorological parameters,

the accuracy of the SED formula, the weight of indicators in the formula and the selected time periods for the study could be considered as the main sources of uncertainties.

Moreover, the most vulnerable provinces and cities are in the western parts of Iran. We should also mention that most of these regions have high unemployment rates and low population growth and economic participation rates, which means that the government has not prioritized the well-being of these regions in the last several decades. These unfavorable statistics originate from the government's biased attention towards megacities, which will probably continue in the future. In this regard, without any adaptation and mitigation policies or clear upstream laws to focus specifically on these regions, climate change will seriously damage residents' lives and worsen the situation in these regions of Iran.

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